

CSV/Excel File Processing System using Django REST Framework

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Subject: Backend Development / Python

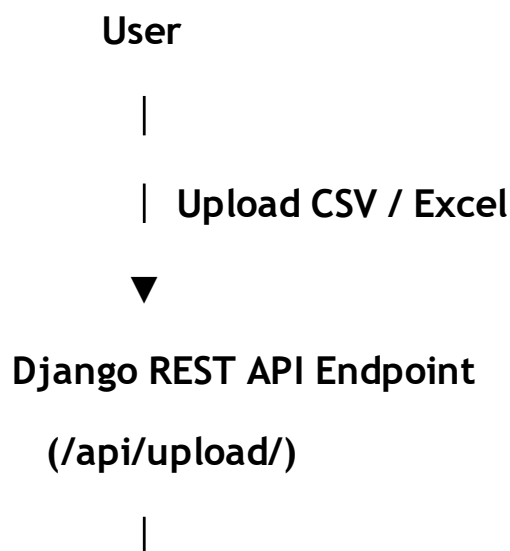
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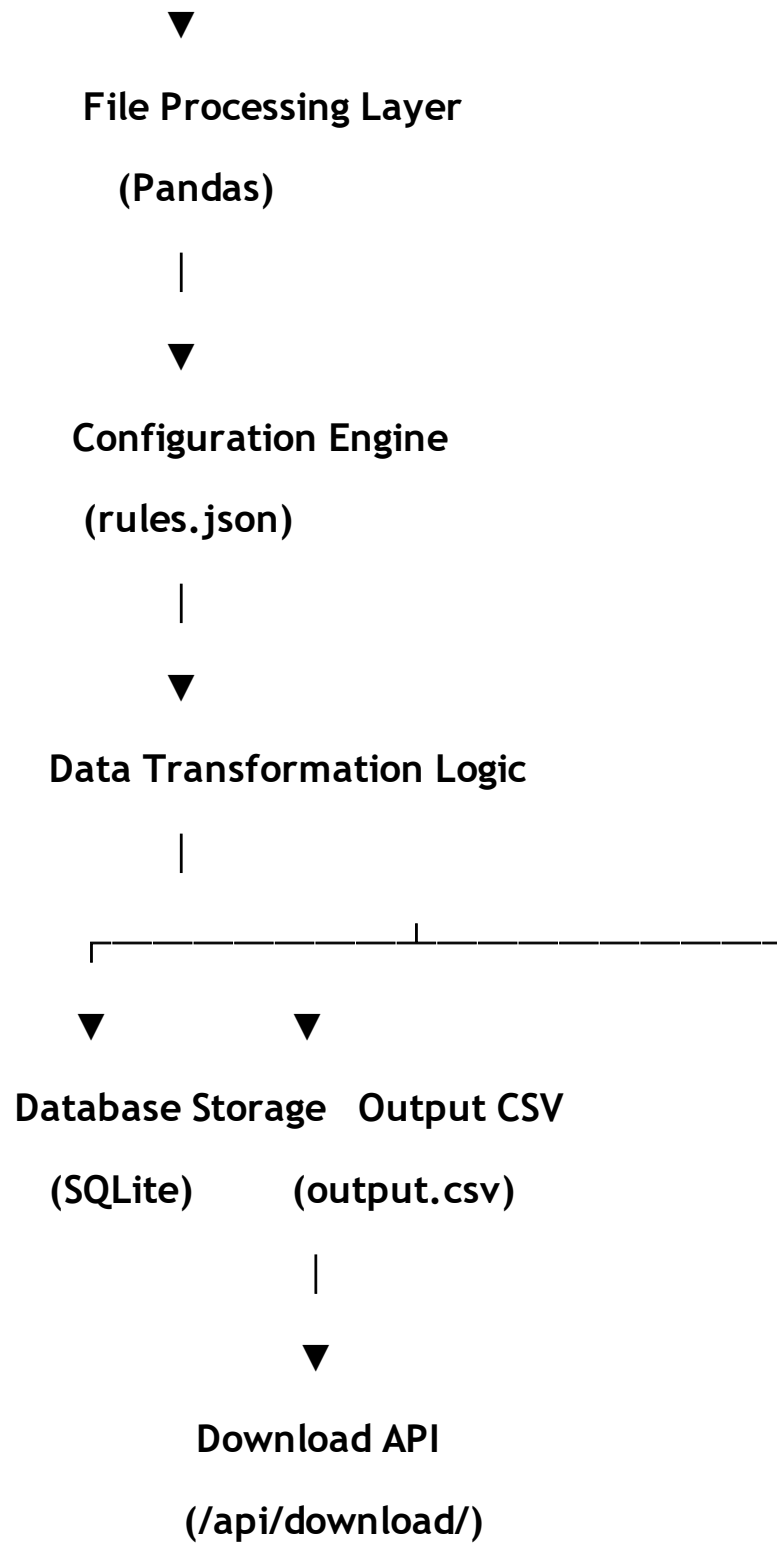
Problem Statement

The goal of this project is to build a backend system using Django and Django REST Framework that can process CSV or Excel files uploaded by users.

The system reads the uploaded file, applies configurable operations defined in a JSON configuration file, and processes the data accordingly. The processed data is stored in a database and exported to a new CSV file.

System Architecture





Project Workflow

1. User uploads CSV or Excel file using API.
2. The system reads the file using Pandas.
3. Configuration rules are loaded from rules.json.
4. Operations are applied to the data.
5. Processed data is stored in the database.
6. Output CSV file is generated.
7. User can download the processed file.

API Documentation

POST /api/upload/

Upload CSV or Excel file

GET /api/download/

Download processed CSV file

Sample Input / Output

Input CSV

```
name,age,salary
Alice,25,50000
Bob,30,60000
Charlie,22,40000
```

Output CSV

```
name,age,salary
Alice,25,60500
Bob,30,71000
```

Technologies Used

Python

Django

Django REST Framework

Pandas

SQLite

JSON

Conclusion

This project demonstrates how a backend system can process CSV or Excel files using Django REST APIs and Pandas. The system supports configuration-based data processing and exports processed data into a new CSV file.